Database Schema: Quiz Management System

April 22, 2026

Table Creation Script

```sql
-- Drop existing tables (order matters due to foreign keys)
DROP TABLE IF EXISTS quiz_responses CASCADE;
DROP TABLE IF EXISTS quiz_questions CASCADE;
DROP TABLE IF EXISTS attendance_log CASCADE;
DROP TABLE IF EXISTS announcements CASCADE;
DROP TABLE IF EXISTS students CASCADE;

-- Create students table
CREATE TABLE students (
    student_id SERIAL PRIMARY KEY,
    username VARCHAR(50) UNIQUE NOT NULL,
    password VARCHAR(100) NOT NULL,
    full_name VARCHAR(100) NOT NULL,
    email VARCHAR(100),
    class_name VARCHAR(50),
    is_active BOOLEAN DEFAULT false,
    last_login TIMESTAMP,
    registered_date TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    role VARCHAR(20) DEFAULT 'student'
);

-- Create attendance_log table
CREATE TABLE attendance_log (
    log_id SERIAL PRIMARY KEY,
    student_id INT REFERENCES students(student_id),
    action VARCHAR(20),
    action_time TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

-- Create announcements table
CREATE TABLE announcements (
    announcement_id SERIAL PRIMARY KEY,
    message TEXT,
    sent_time TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

-- Create quiz_questions table
CREATE TABLE quiz_questions (
    question_id SERIAL PRIMARY KEY,
    question_text TEXT,
    option_a VARCHAR(500),
    option_b VARCHAR(500),
    option_c VARCHAR(500),
    option_d VARCHAR(500),
```

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    correct_answer CHAR(1),
    created_time TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

-- Create quiz_responses table
CREATE TABLE quiz_responses (
    response_id SERIAL PRIMARY KEY,
    student_id INT REFERENCES students(student_id),
    question_id INT REFERENCES quiz_questions(question_id),
    student_answer CHAR(1),
    is_correct BOOLEAN,
    submitted_time TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

-- Insert a test student (optional)
INSERT INTO students (username, password, full_name, class_name, role)
VALUES ('teststudent', '5
    e884898da28047151d0e56f8dc6292773603d0d6aabbdd62a11ef721d1542d8',
        'Test Student', 'Class A', 'student');

-- Verify everything
SELECT * FROM students;
```

Listing 1: Database Schema SQL Script

Table Descriptions

- **students**: Stores user account information and profile data
- **attendance_log**: Tracks student attendance actions
- **announcements**: Stores system announcements
- **quiz_questions**: Contains quiz questions and answer choices
- **quiz_responses**: Records student answers to quiz questions

Entity Relationship Notes

- Foreign key relationships exist between quiz_responses and both students and quiz_questions
- attendance_log references students for attendance tracking
- Cascade deletes ensure referential integrity when dropping tables